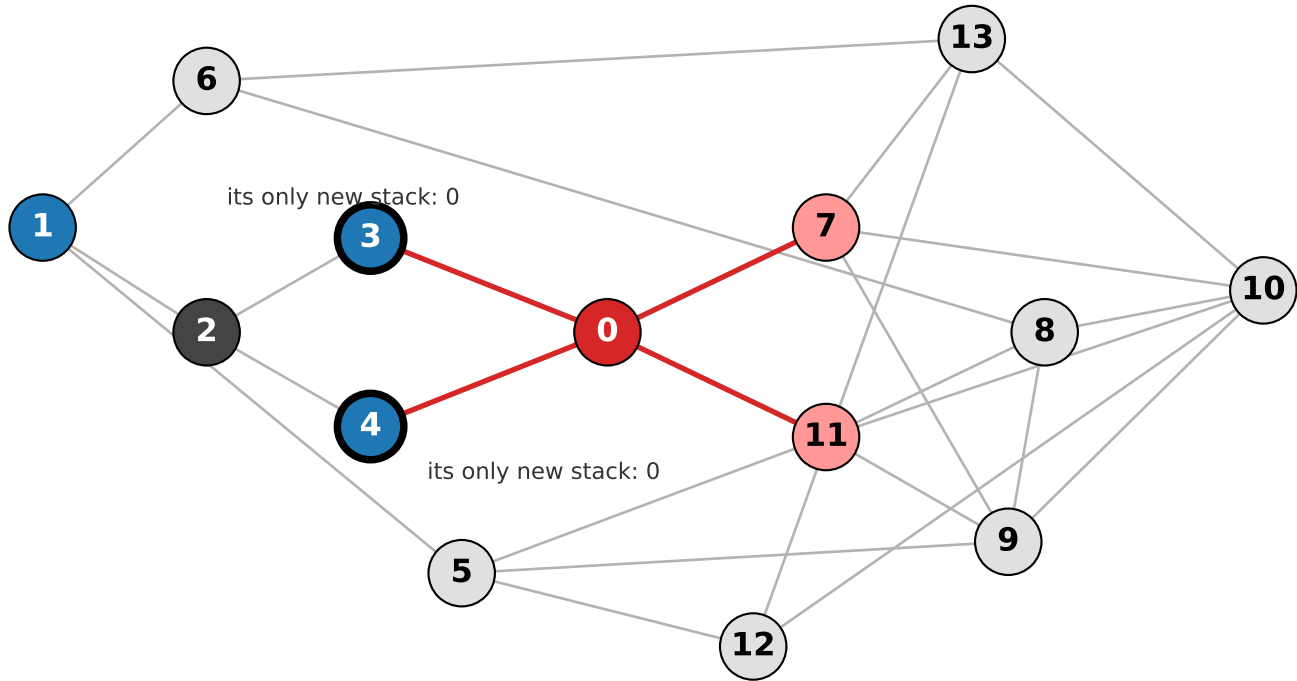


A counterexample to Chu & Stuckey's Theorem 1, at $k = 6$
 0 is playable and $\text{close}(0, S) \geq \text{open}(0, S)$, so the rule keeps 0 alone. A solution from S exists
 (first moves 1, 3, 4, 6), but none starts with 0.



- closed: $S = \{2\}$
- open after S : 1, 3, 4
- $q = 0$, the move the definite move keeps
- new stacks q opens: 0, 7, 11, 0 itself included (open = 3)
- other customers
- $\text{close}(0, S)$ counts 0, 3, 4; 3 and 4 share one new stack